Karli optimization problem

The following document describes the current status of the Karli optimization algorithm.

Variables

In the following, all variables are explained.

Continous variables

The following variables are continous and nonnegative.

Variable	Description
$\epsilon^{\text{out}}_{\text{g,t}}$	Grid consumption (electricity going *out* of the grid)
$\epsilon^{\text{in}}_{\text{g,t}}$	Grid feed-in (electricity going *in* to the grid)
$\epsilon^{\text{out}}_{\text{s,t}}$	Battery discharge (electricity going *out* of the storage)
$\epsilon^{\text{in}}_{\text{s,t}}$	Battery charge (electricity going *in* to the storage)
$\epsilon^{\text{con}}_{\text{s,t}}$	Battery storage content

Binary variables

The following variables only have the states $\{0, 1\}$.

Variable	Description
$\beta_{\text{g,t}}$	Feed-in vs. consumption
$\beta_{\text{s,t}}$	Charge vs. discharge

Parameters

The following tables contain the constant parameters for the equations.

Parameter	Description
t	Time step
T	Time interval (15 minutes mostly)
$c^{\text{out}}_{\text{g,t}}$	Grid consumption price / electricity costs
$c^{\text{in}}_{\text{g,t}}$	Feed-in revenue / feed-in tariff
$e_{\text{PV,t}}$	PV production
$e_{\text{con,t}}$	Household consumption
$k^{\text{min}}_{\text{s}}$	Minimum storage capacity

Parameter	Description
$k_{\mathrm{s}}^{\mathrm{max}}$	Maximum storage capacity
$k_{\mathrm{s}}^{\mathrm{inv}}$	Inverter power
$r_{\mathrm{s}}^{\mathrm{out}}$	Discharge efficiency
$r_{\mathrm{s}}^{\mathrm{in}}$	Charge efficiency
M	"Big M" slack penalty

Objective

The objective function of the Karli optimization problem reads as follows:

$$\min \sum^{\mathrm{t}} c_{\mathrm{g,t}}^{\mathrm{out}} \cdot \epsilon_{\mathrm{g,t}}^{\mathrm{out}} - c_{\mathrm{g,t}}^{\mathrm{in}} \cdot \epsilon_{\mathrm{g,t}}^{\mathrm{in}}$$

Constraints

The variables are limited by the following constraints.

Vertex rule

All flows into the grid connection point (GCP) and out of it must add up:

$$\epsilon_{\mathrm{g,t}}^{\mathrm{out}} + \epsilon_{\mathrm{s,t}}^{\mathrm{out}} - \epsilon_{\mathrm{g,t}}^{\mathrm{in}} - \epsilon_{\mathrm{s,t}}^{\mathrm{in}} + e_{\mathrm{PV,t}} - e_{\mathrm{con,t}} = 0$$

Storage limitations

The storage content is limited by the storage capacities.

$$k_{\mathrm{s}}^{\mathrm{min}} \leq \epsilon_{\mathrm{s,t}}^{\mathrm{con}} \leq k_{\mathrm{s}}^{\mathrm{max}}$$

The storage content is dependent upon the charge and discharge and its former state.

$$\epsilon_{\mathrm{s,t}}^{\mathrm{con}} = \epsilon_{\mathrm{s,t\text{-}1}}^{\mathrm{con}} + \epsilon_{\mathrm{s,t}}^{\mathrm{in}} \cdot r_{\mathrm{s}}^{\mathrm{in}} - \epsilon_{\mathrm{s,t}}^{\mathrm{out}} \cdot \frac{1}{r_{\mathrm{s}}^{\mathrm{out}}}$$

The charge and discharge are limited by the inverter power:

$$\epsilon_{\mathrm{s,t}}^{\mathrm{in}} \leq k_{\mathrm{s}}^{\mathrm{inv}} \cdot T$$

$$\epsilon_{\mathrm{s,t}}^{\mathrm{out}} \leq k_{\mathrm{s}}^{\mathrm{inv}} \cdot T$$

The storage should not be charged and discharged at the same time:

$$\epsilon_{\mathrm{s,t}}^{\mathrm{out}} \leq M \cdot \beta_{\mathrm{s,t}}$$

$$\epsilon_{\mathrm{s,t}}^{\mathrm{in}} \leq M \cdot (1 - \beta_{\mathrm{s,t}})$$

Grid limitation

Grid feed-in cannot exceed the PV production:

$$\epsilon_{\mathrm{g,t}}^{\mathrm{in}} \leq e_{\mathrm{PV,t}}$$

Only grid consumption **or** grid feed-in can happen in one time step:

$$\epsilon_{\mathrm{g,t}}^{\mathrm{out}} \leq M \cdot \beta_{\mathrm{g,t}}$$

$$\epsilon_{\mathrm{g,t}}^{\mathrm{in}} \leq M \cdot (1 - \beta_{\mathrm{g,t}})$$